



HospitalIQ

End-to-End Healthcare Analytics & ML Platform

SQL Server · Power BI · Python/XGBoost · SHAP

27,891 encounters, 974 patients, \$101.5M in claims, AUC 0.907

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Executive Summary



An end-to-end hospital analytics project on synthetic EHR data (Synthea) — mirroring real production methodology: schema design, BULK INSERT loading, rigorous data quality auditing, multi-layered analysis, and explainable ML.

27,891

Encounters

974

Patients

\$101.5M

Total Claims

0.907

Model AUC

Headline Findings

Encounter Volume

Peaked in 2014 (3,885) and 2021 (3,530). Outpatient surged to 40.2% in 2021 — likely a COVID-era coding shift.

Payer Coverage Gap

48.7% of encounters had zero payer coverage. 4,779 were insured patients (deductibles or denied claims) — a nuance most reporting misses.

Cost Concentration

Electrical cardioversion drives ~\$35.8M (1,383 procedures × ~\$25,903). ICU admissions rare (n=5) but cost \$206K each on average.

Payer Disparities

Average claim cost varies 3.7× across payers. Medicaid (\$6,205) and uninsured (\$5,593) carry the highest costs — consistent with delayed-care patterns.

Readmission Model

XGBoost achieves AUC 0.907 predicting 30-day readmission, driven by clinical features (prior encounter count, recency, claim cost, age) — not demographics.

Project Phases

1

Data Engineering

Schema, BULK INSERT, data quality

2

SQL Analytics

10 business-question queries

3

Power BI Dashboard

3 pages: Operations, Financial, Patient Risk

4

Python ML Model

30-day readmission, XGBoost + SHAP

All four phases completed on a Synthea synthetic EHR dataset (974 patients, 27,891 encounters, 47,701 procedures, spanning 2011–2022).

Data Engineering & Quality

Loading Issues Resolved

1 Missing Patient Row

BULK INSERT silently skipped one record (19 dependent encounters) due to special characters. Detected via LEFT JOIN orphan check; resolved with SSMS Import Flat File Wizard.

2 Empty Strings vs NULLs

Optional fields loaded as "" instead of NULL, making every patient appear alive. Fixed with NULLIF() — 154 patients with valid death dates, exactly matching source CSV.

Findings Documented, Not "Fixed"

70% NULL reason codes

Routine visits have no diagnosis — left as NULL.

53 post-death encounters

Admin/billing entries — kept for cost analysis; excluded from patient behavior.

1 patient, 1,381 encounters

Chronic-care profile (likely dialysis) — flagged to prevent readmission model misinterpretation.

4,779 zero-coverage encounters

Insured patients with deductibles or denied claims — both interpretations reported explicitly.

SQL Analytics: Key Findings

Encounter Volume

Range: 220 (2022, partial) to 3,885 (2014). Ambulatory 37–60% every year. 95.87% of encounters under 24 hours.

Zero Coverage

13,586 encounters (48.71%). The gap of 4,779 insured patients with \$0 coverage reflects deductibles or denied claims — both numbers reported.

Top Costs

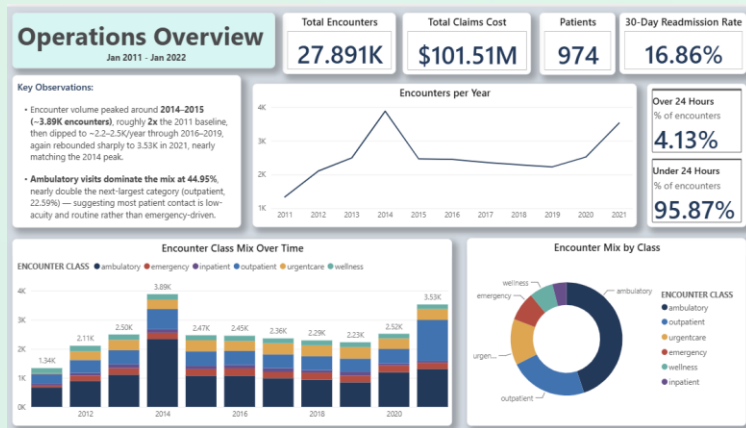
ICU admit (\$206K, n=5), CABG (\$47K, n=9), Hemodialysis (\$29K, n=27). Medicaid \$6,205 avg vs Medicare \$2,167 — a 3.7× range.

Readmissions

299 patients (79.3%) readmitted within 30 days (broad definition); 29 under CMS-aligned inpatient-only definition. Top readmitter: Kimberly Collier, 1,375 readmissions.

Power BI Dashboard

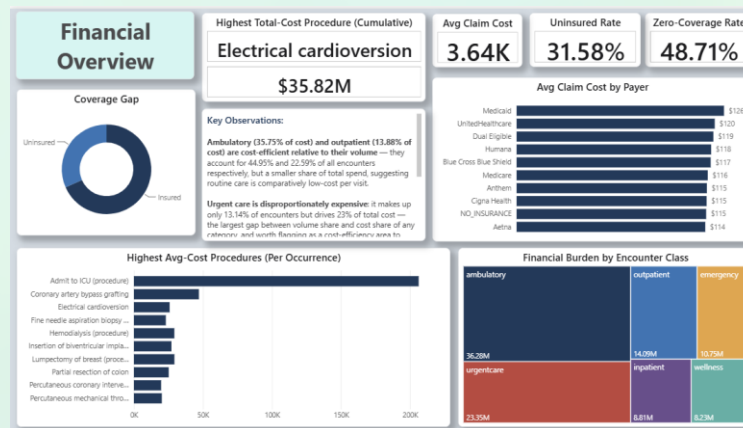
Three-page dashboard in Import mode on a star schema with *encounters* as fact table and four dimensions (patients, payers, organizations, procedures). All measures DAX-defined. Embedded with cached data — no live SQL Server connection needed



1

Operations Overview

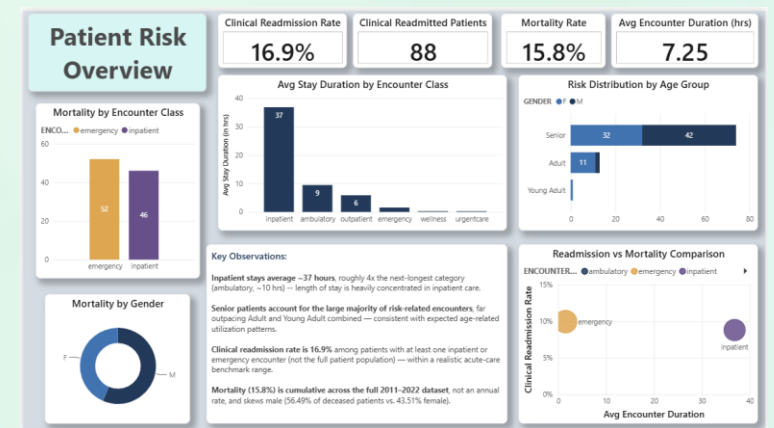
KPI cards, year-over-year trend, encounter-class donut, and class-mix stacked bar. 2014 and 2021 volume peaks visible at a glance.



2

Financial Overview

Average claim cost by payer, coverage-type donut, top-10 procedures by cost, and financial burden treemap by encounter class.



3

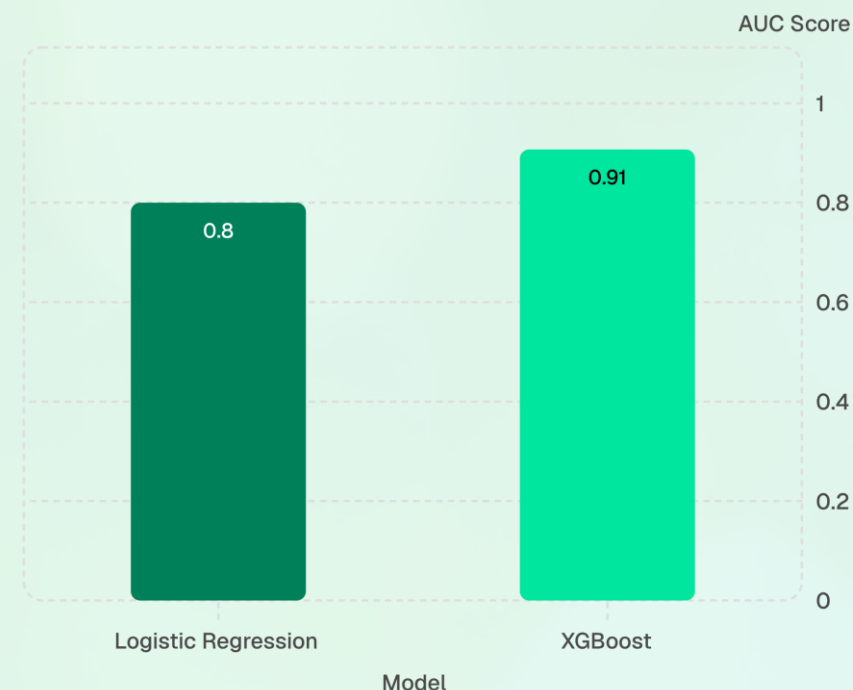
Patient Risk & Readmission

Clinical readmission rate (strict definition), mortality rate, average stay duration, and risk distribution by age and gender.

Readmission Prediction Model

Approach

- **Feature engineering in SQL** - LAG, LEAD, COUNT OVER window functions compute prior encounter count and days since previous visit
- **Two models compared** - Logistic Regression baseline vs XGBoost
- **80/20 stratified split** - 22,312 train / 5,579 test, identical 61.6% target distribution
- **SHAP explainability** - summary and beeswarm plots on 500 test predictions



Real-world EHR readmission models typically achieve AUC 0.65–0.75. The higher score here reflects Synthea's cleaner patterns - methodology transfers directly.

The 10.7-point AUC gap justifies XGBoost's additional complexity over the interpretable baseline.

Feature Importance (SHAP)

→ prior_encounter_count

More prior encounters → higher readmission probability

→ days_since_prev

Recent prior visit → higher readmission probability

→ claim_cost

Higher cost (sicker patient) → higher readmission probability

→ patient_age

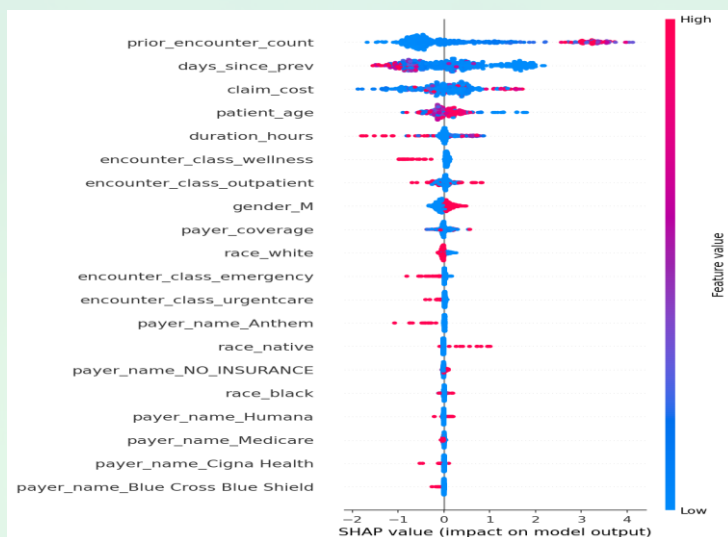
Older patient → modestly higher readmission probability

→ duration_hours

Longer encounter → higher readmission probability

→ encounter_class_wellness

Wellness check → pushes away from readmission



Clinical & Behavioral Features Dominate

The model's top predictors are clinically meaningful — exactly what a credible healthcare model should look like. Race and gender appear near the bottom of the importance ranking.

Conclusions & Next Steps

Coverage Gap

35% of zero-coverage encounters belonged to insured patients. Track both metrics — "uninsured rate" alone under-counts those exposed to large out-of-pocket costs.

Cost Concentration

One procedure (electrical cardioversion) drives ~\$35M. Prioritize cost management on high-impact procedures rather than spreading thinly across all encounter types.

Preventative Outreach

Medicaid and uninsured patients carry the highest average costs - consistent with delayed-care patterns. Targeted outreach could reduce long-term costs more effectively than acute-care interventions alone.

Next Steps

Add chronic-care stratification; calibrate probabilities (Platt scaling); replicate on real EHR data via NHS or US health system partnership; wrap model in REST API for discharge-planning UI.

Data source: Walonoski J, Kramer M, Nichols J, et al. *Synthea: An approach, method, and software mechanism for generating synthetic patients and the synthetic electronic health care record*. JAMIA, March 2018.